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Comparative safety and health service utilization for gabapentin versus non-steroidal anti-inflammatory drugs for cervical radiculopathy: A retrospective cohort study of academic centers

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ABSTRACT

Background: Off-label use of gabapentin for neuropathic pain conditions such as cervical radiculopathy (CR) remains controversial due to limited evidence of benefit and safety concerns (e.g., opioid co-prescription, adverse drug events [ADEs], dizziness, and hospitalization risk), necessitating further research. This study explores safety risks and health service utilization in adults with newly diagnosed CR, comparing gabapentin to nonsteroidal anti-inflammatory drugs (NSAIDs) to inform clinical practice.

Methods: This retrospective propensity-matched cohort study using the TriNetX database included opioid-naïve adults (≥18 years) with newly-diagnosed CR without history of spinal surgery, spinal neoplasms, myelopathy, opioid-use disorder or cervical injection. Patients were divided into gabapentin or NSAID cohorts based on prescription at CR diagnosis, and matched for demographics, prescriptions, and comorbidities. Outcomes included incidence and risk ratios (RRs) of safety and health service utilization markers assessed through one year. Significance was evaluated at $P < 0.0036$ (Bonferroni-corrected).

Results: After matching, there were 23,379 patients (mean age: 53 years; 57% female) per cohort. About 14% of patients per cohort were co-prescribed opioids at CR diagnosis. Compared to the NSAID cohort, the gabapentin cohort had a statistically significant ($P < 0.0036$) increased risk of posterior cervical fusion (RR: 3.23), opioid-related disorder (RR: 2.87), cervical corticosteroid injection (RR: 2.52), opioid-related ADEs (RR: 2.32), anterior cervical discectomy and fusion (RR: 2.10), acute respiratory failure (RR: 1.80), cervical spine magnetic resonance imaging (RR: 1.43), inpatient visits (RR: 1.38), oral opioid prescription (RR: 1.34), and cervical spine radiographic imaging (RR: 1.21). Risk of dizziness, falls, somnolence, and emergency department visits was not significantly increased ($P > 0.0036$).

Conclusion: Gabapentin prescription for newly-diagnosed CR in adults was associated with increases in opioid prescription, adverse safety events, and health service utilization compared to NSAIDs, corroborating its lack of guideline recommendation for CR. No outcome favored gabapentin over NSAIDs. Gabapentin was occasionally co-prescribed with opioids at CR diagnosis, raising concerns.

Level of Evidence: Level IV.

1. Introduction

Gabapentin is an antiepileptic drug with analgesic properties that is commonly used as an off-label non-opioid alternative for neuropathic pain [1–3]. Although working through various mechanisms, gabapentin primarily targets the α2δ-1 subunit of neural voltage gated calcium

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channels to attenuate neuropathic pain [4]. Despite limited data regarding its efficacy and emerging evidence of potential harms, gabapentin is frequently prescribed for a variety of musculoskeletal and neuropathic pain conditions [5]. Cervical radiculopathy (CR), a disorder involving cervical nerve root compression with associated pain and/or deficits, is one such condition. In a recent retrospective United States (US) study of 12,514 patients with CR who underwent anterior cervical spine surgery, 18% received gabapentin in the year preceding surgery [6]. Despite its relatively common use, limited research has explored the clinical impact and incidence of potential adverse drug events (ADEs) potentially associated with gabapentin in patients with CR. While several studies have assessed the safety and efficacy of gabapentin prescription for lumbar radiculopathy [7–12], little data exists on associated health service utilization.

Importantly, CR is often self-limiting, with over half of patients recovering within six to 12 months [13]. Accordingly, guideline recommendations for CR prioritize non-surgical interventions including patient education, exercise, manual therapy, traction, with careful consideration of nonsteroidal anti-inflammatory drugs (NSAIDs) or tramadol, in the absence of myelopathy or serious pathology [14]. Notably, gabapentin is not recommended in any clinical practice guideline for CR to our knowledge [15]. Currently in the US, the only pain-related Food and Drug Administration (FDA) approved indication for gabapentin is postherpetic neuralgia [16]. Given the limited evidence supporting its utility and its potential harms, multiple authors have critiqued routine gabapentin prescribing practices, urging cessation of its off-label prescription [3,17,18], including a recent article published by Rickert (2025) in Clinical Orthopaedics and Related Research (CORR) [5]. Spurring such concerns, outpatient gabapentin prescriptions has quadrupled from 2003 to 2016 [19], with 90% to 99% of prescriptions being for off-label indications [5,19,20], suggesting limited real-world adherence to clinical practice guidelines and FDA prescribing indications. Additionally, primary care physicians, neurologists, and surgeons, who often encounter musculoskeletal disorders such as CR, are the most frequent prescribers of gabapentin [19].

Gabapentin increases the risk of a spectrum of ADEs, many linked to its mechanism of central nervous system depression. A randomized trial focusing on CR in which 66 patients received gabapentin found 45.4% of these patients experienced ADEs, including cognitive impairment, fatigue, and others [21]. Broader pharmacovigilance databases suggest somnolence, confusion, dizziness, hallucinations, and agitation are other potential related ADEs [22], with older adults taking gabapentin being more at risk of falls [5]. Risks of opioid-related ADEs increase when gabapentin is co-prescribed with opioids, including overdose and respiratory depression [23–25]. Gabapentin is also associated with increased healthcare utilization, including inpatient and emergency department visits [26]. Furthermore, in outpatient settings, one third of gabapentin prescriptions occur concurrently with opioid prescriptions despite evidence that this increases the risk of ADEs [19]. In general, inappropriate opioid prescription for newly diagnosed musculoskeletal conditions remains commonplace [27]. Considering the common use of gabapentin and opioids for conditions like CR, a large-scale assessment of potential ADEs and broader health service utilization is needed at this time to shine additional light on clinician prescription practices.

Therefore, the purpose of this study is to explore the incidence and risk ratio (RR) of safety markers (e.g., opioid prescription & related ADEs) and health service utilization markers (e.g., cervical spine surgery, cervical corticosteroid injections and emergency/inpatient visits), comparing patients who were prescribed gabapentin versus nonsteroidal anti-inflammatory drugs (NSAIDs) at the index date of a new diagnosis of CR through a one-year follow-up window. This analysis seeks to provide clinicians with clearer evidence on the safety profile and healthcare implications of gabapentin in management of CR.

2. Methods

2.1. Study registration and design

This retrospective propensity-matched cohort study adhered to an a priori protocol via Open Science Framework (components on https://osf.io/hym3v/) with one modification described below in our power analysis to use a more conservative, Bonferroni-adjusted alpha as the original analysis did not correct for multiple tests. This study was exempt from Institutional Review Board (IRB) review and patient consent requirements as the institutional IRB considers study designs which use de-identified data from TriNetX (TriNetX, LLC., Cambridge, MA, US) in accordance with standard operating procedures to meet criteria for "Not Human Subjects" research. The query date for this cohort study was on March 28th, 2025.

2.2. Data source and study setting

Data were obtained from TriNetX, a large online research network based on the electronic health records of 105 large healthcare organizations with 145 million patients, including medical centers and their affiliated outpatient offices. TriNetX uses International Classification of Disease (ICD) codes and Current Procedural Terminology codes (CPT), among others (e.g., RxNorm for medications). Furthermore, TriNetX is compliant with the Health Insurance Portability and Accountability Act (HIPAA). Data displayed on the TriNetX Platform are in aggregate form and de-identified as per the standard defined in Section §164.514(a) of the HIPAA Privacy Rule, and participating organizations remain anonymous.

2.3. Study participants

Patients were adults (≥18 years old) with a first-time ICD-10-code-based diagnosis of CR between 2004 and 2024, at least one year prior to the query for follow-up, and notably defined without a formal diagnostic look-back period. Patients were excluded if they had any history of spine surgery, cervical corticosteroid injection, spinal neoplastic conditions, cervical myelopathy, opioid related disorder, or any conditions for which gabapentinoids are FDA approved: postherpetic neuralgia [16,28]), epilepsy [28], spinal cord injury [16], fibromyalgia [16], diabetic neuropathy [16], and restless leg syndrome [29]. Furthermore, as seen in a prior study [30], contraindications for gabapentin were excluded (i.e., myoclonus and myasthenia gravis) [31]. To help standardize cohorts to receiving new care for CR and establish baseline comparability, we excluded those with an opioid prescription within the year prior to CR diagnosis. To minimize loss to follow-up, both cohorts were required to have a healthcare visit of any kind during the one-year follow-up window. Patients were divided into two cohorts based on the prescription of gabapentin (gabapentin cohort; Supplemental Table 1) or NSAIDs (NSAID cohort; Supplemental Table 2) on the day of first diagnosis of CR. NSAIDs were chosen as an active comparator as these are recommended by clinical practice guidelines for CR [14]. Additionally, patients in the NSAID cohort who received gabapentin through one year were excluded to minimize crossover between cohorts, resulting in an approximate 26% reduction in this cohort's initial size. Accordingly, it should be noted that the present study-participant-design best generalizes to an initial prescribing strategy, rather than sustained treatments or care pathways following the index date.

2.4. Propensity matching variables

To reduce confounding, we propensity-matched on risk factors associated with CR complexity, gabapentin prescription, and opioid prescription, assessed one year preceding and including the date of CR diagnosis. These included demographics (age, sex, ethnicity/race) [32–35], diagnoses of chronic pain [33,36], complex regional pain

syndrome [36], chronic obstructive pulmonary disease [35], adverse socioeconomic factors [19,33,34], diabetes [37], emergency and inpatient visits [38], headaches and migraines [33], insomnia [38], irritable bowel syndrome [38], mood disorders [35,39], anxiety disorders [35,39], mental and behavioral disorders related to psychoactive substance use [37,38], prescription of adjuvant analgesics (i.e., antiarrhythmics, antidepressants, benzodiazepines, corticosteroids, muscle relaxants, serotonin-norepinephrine reuptake inhibitors, other anticonvulsants (i.e., topiramate, pregabalin), tricyclic antidepressants) [33,34,36], and tobacco use/dependence [19,35,36]. We also matched on cervical stenosis and the three inclusionary CR diagnoses to ensure cohort similarity.

2.5. Outcomes

The outcomes for this retrospective cohort study explored the incidences and risk ratios (RRs) of a range of safety and health service utilization events occurring between one day and one year following first-time CR diagnosis (Supplemental Table 3). Safety and ADE outcomes encompass oral opioid prescription, opioid-related disorders, opioid-related ADEs (e.g., overdose & poisoning), acute respiratory failure, dizziness, somnolence/disorientation, and falls. These were selected due to their association with gabapentin in prior studies [21–24], in addition to the warnings in a recent letter by Rickert (2025) [5]. Health service utilization outcomes include anterior cervical discectomy and fusion (ACDF), posterior cervical fusion (PCF), cervical corticosteroid injection, emergency department (ED) visits, inpatient visits, cervical spine radiographs, and cervical magnetic resonance imaging (MRI), which are justified by the evidence of increased healthcare linked to off-label prescription of gabapentin [26].

2.6. Power analysis for sample size

A required sample size of 23,304 patients per cohort (46,608 in total) was calculated using previous data regarding respiratory depression among gabapentin users [26], as to adequately power the study for rarer outcomes rather than more common ones such as cervical imaging. We powered the study to detect a difference of 0.2% incidence between cohorts (gabapentin cohort 0.4% vs. NSAID cohort: 0.2%) [26]. This updated power analysis used a Bonferroni-corrected two-tailed alpha-error of 0.0036, an allocation ratio of one, and a power of 0.85 using GPower (Kiel University, DE) with the z-testing stats function.

2.7. Statistical analysis

The statistical analysis was completed using the TriNetX online statistical software and R (version 4.2.2, Vienna, AT) [40]. For comparisons of patient baseline characteristics, this study used standardized mean difference (SMD), with a threshold of SMD > 0.1 to represent meaningful baseline asymmetry. TriNetX uses Python (scikit-learn version 1.3 [Python Software Foundation, Delaware, USA]) to conduct logistic regression for calculation of propensity scores. To explore data quality, we reported the mean and median follow-up duration, as well as the proportion of patients having at least one year follow-up. We calculated RRs with 95% confidence intervals (CIs) for each outcome, using the Bonferroni correction to adjust the statistical significance threshold to $p < 0.0036$ for the 14 exploratory outcomes, while visualizing effect estimates via a forest plot. We also calculated absolute risk increase (ARI) for statistically significant results. We used R to calculate 95% CIs and ggplot2 [41] to graph RRs, cumulative incidences, and propensity score densities. We also calculated the RR for negative control outcomes to assess for differences in healthcare engagement or documentation bias between cohorts, evaluating the RR point estimates around the null as balanced (i.e., values of 0.8 to 1.2) [42]. For negative control outcomes unrelated to gabapentin prescription, we calculated RRs for colonoscopy, tetanus, diphtheria, and acellular pertussis vaccination (TDAP), as well as urinary tract infection (Supplemental Table 4).

3. Results

3.1. Baseline patient characteristics

Prior to propensity matching, there were 25,717 patients in the gabapentin cohort and 39,302 patients in the NSAID cohort. After propensity matching, there were 23,379 patients in both the gabapentin and NSAID cohorts, respectively. Prior to matching, patients in the gabapentin cohort a greater mean age at time of CR diagnosis and proportion of white individuals, among other differences (SMD > 0.1). After propensity matching, the two cohorts were similar in terms of all matched baseline characteristics (SMD < 0.1)(Table 1). Although we avoided matching on opioid prescription at baseline to allow us to observe real-world prescription patterns for CR, both the gabapentin cohort and the NSAID cohort had similar rates of opioid co-prescription on the date of CR diagnosis (SMD $= 0.01$; 14.3% versus 13.9%).

3.2. Data quality and propensity matching diagnostics

There was an adequate amount of information (>1000 average facts per patient) for patients in both cohorts. For the gabapentin cohort, the follow-up duration was a median (interquartile range [IQR]) of 365 (0) days and a mean (SD) of 315.1 (106.4) days with 77.8% of patients having complete follow-up. Similarly, for the NSAID cohort, the follow-up duration was a median [IQR] of 365 (0) days and a mean (SD) of 315.4 (107.0) days with 78.1% of patients having complete follow-up. Cohorts were balanced in terms of healthcare engagement as the negative control outcome effect sizes for TDAP vaccination (RR: 0.95), urinary traction infection (RR: 0.93), and colonoscopy (RR: 0.99) were around the null (RR: 0.8–1.2). Visual representation of propensity score densities between both cohorts were also nearly indistinguishable after propensity matching, further suggesting the cohorts were adequately similar in terms of baseline characteristics (Fig. 1). These observations suggest that our results were unlikely to be related to differences in patient follow-up, data density, documentation bias, or healthcare engagement.

3.3. Markers of safety and adverse drug events for gabapentin versus NSAIDs

As compared to patients in the NSAID cohort, patients in the gabapentin cohort had a statistically significant higher risk of acute respiratory failure ($p < 0.001$; RR: 1.80 [1.35, 2.39]; ARI: 0.3%; 0.6% versus 0.3%), opioid-related ADEs ($p < 0.001$; RR: 2.32 [1.49, 3.61]; ARI: 0.2%; 0.3% versus 0.1%), oral opioid prescription ($p < 0.001$; RR: 1.34 [1.28, 1.39]; ARI: 4.9%; 19.2% versus 14.3%; Fig. 2), and new diagnosis of opioid-related disorders ($p < 0.001$; RR: 2.87 [1.91, 4.32]; ARI: 0.3%; 0.4% versus 0.1%) through one year after first-time diagnosis of CR (Table 2; Fig. 3). However, there was no statistically significant difference in risk of dizziness ($p = 0.110$; RR: 1.08 [0.98, 1.18]; 4.1% versus 3.8%), falls ($p = 0.108$; RR: 1.12 [0.98, 1.29]; 1.7% versus 1.5%), or somnolence and/or disorientation ($p = 0.092$; RR: 1.27 [0.96, 1.67]; 0.5% versus 0.4%).

3.4. Markers of health service utilization for gabapentin versus NSAIDs

As compared to patients in the NSAID cohort, patients in the gabapentin cohort had a statistically significant higher risk of ACDF ($p < 0.001$; RR: 2.10 [1.81, 2.44]; ARI: 1.2%; 2.3% versus 1.1%), PCF ($p < 0.001$; RR: 3.23 [2.16, 4.82]; ARI: 0.3%; 0.4% versus 0.1%), cervical corticosteroid injection ($p < 0.001$; RR: 2.52 [2.30, 2.77]; ARI: 3.8%; 6.3% versus 2.5%), cervical spine MRI ($p < 0.001$; RR: 1.43 [1.38, 1.49]; ARI: 6.3%; 20.8% versus 14.5%), cervical spine radiographic imaging ($p < 0.001$; RR: 1.21 [1.15, 1.27]; ARI: 2.0%; 12.0% versus 10.0%), and

Table 1
Baseline patient characteristics before and after propensity matching.

Variable (n (%) or mean (SD[a]))	Before matching Gabapentin cohort	NSAID[b] cohort	SMD[c]	After matching Gabapentin cohort	NSAID cohort	SMD
Patients (n)	25,717	39,302	–	23,379	23,379	–
Cervical radiculopathy (M54.12)	23,094 (90%)	33,306 (85%)	**0.152**	20,831 (89%)	20,939 (90%)	0.015
Cervical spondylosis with radiculopathy (M47.22)	2448 (10%)	5117 (13%)	**0.111**	2269 (10%)	2146 (9%)	0.018
Cervical disc disorder with radiculopathy (M50.1)	1707 (7%)	2475 (6%)	0.014	1489 (6%)	1492 (6%)	0.001
Current age (years)	59.5 (13.5)	55.9 (14.2)	**0.257**	58.7 (13.4)	58.6 (13.9)	0.006
Age at index (years)	54.0 (13.3)	50.1 (13.9)	**0.285**	53.1 (13.1)	53.0 (13.5)	0.008
White	16,535 (64%)	22,810 (58%)	**0.129**	14,795 (63%)	14,939 (64%)	0.013
Female	14,634 (57%)	23,073 (59%)	0.037	13,406 (57%)	13,382 (57%)	0.002
Male	11,066 (43%)	16,202 (41%)	0.037	9957 (43%)	9979 (43%)	0.002
Black or African American	3821 (15%)	6560 (17%)	0.05	3538 (15%)	3556 (15%)	0.002
Hispanic or Latino	1662 (6%)	2966 (8%)	0.042	1549 (7%)	1545 (7%)	0.001
Asian	938 (4%)	1466 (4%)	0.004	873 (4%)	873 (4%)	0
Other chronic pain	3868 (15%)	4145 (11%)	**0.135**	3218 (14%)	3190 (14%)	0.003
Anxiety, dissociative, stress-related, somatoform and other nonpsychotic mental disorders	3009 (12%)	4073 (10%)	0.043	2629 (11%)	2703 (12%)	0.01
Diabetes mellitus	2375 (9%)	2859 (7%)	0.071	2026 (9%)	1980 (8%)	0.007
Cervical spinal stenosis	2698 (10%)	2072 (5%)	**0.195**	1961 (8%)	1810 (8%)	0.024
Mental and behavioral disorders due to psychoactive substance use	1954 (8%)	3048 (8%)	0.006	1727 (7%)	1695 (7%)	0.005
Nicotine dependence	1647 (6%)	2701 (7%)	0.019	1473 (6%)	1433 (6%)	0.007
Headache	1353 (5%)	2311 (6%)	0.027	1244 (5%)	1291 (6%)	0.009
Migraine	1129 (4%)	1473 (4%)	0.033	995 (4%)	1003 (4%)	0.002
Personal history of nicotine dependence	829 (3%)	1130 (3%)	0.02	710 (3%)	708 (3%)	0
Insomnia	811 (3%)	954 (2%)	0.044	690 (3%)	683 (3%)	0.002
Chronic obstructive pulmonary disease	598 (2%)	567 (1%)	0.065	459 (2%)	443 (2%)	0.005
Irritable bowel syndrome	319 (1%)	389 (1%)	0.024	271 (1%)	262 (1%)	0.004
Persons with potential health hazards related to socioeconomic and psychosocial circumstances	172 (1%)	209 (1%)	0.018	147 (1%)	141 (1%)	0.003
Persistent mood [affective] disorders	152 (1%)	206 (1%)	0.009	132 (1%)	129 (1%)	0.002
Personal history of other mental and behavioral disorders	56 (<1%)	81 (<1%)	0.003	49 (<1%)	48 (<1%)	0.001
Complex regional pain syndrome	47 (<1%)	27 (<1%)	0.032	29 (<1%)	25 (<1%)	0.005
Skeletal muscle relaxants	7914 (31%)	17,389 (44%)	**0.281**	7676 (33%)	7522 (32%)	0.014
Antidepressants	5834 (23%)	6191 (16%)	**0.177**	4845 (21%)	4821 (21%)	0.003
Corticosteroids	5320 (21%)	8262 (21%)	0.008	4831 (21%)	4848 (21%)	0.002
Opioids*	3650 (14%)	6091 (15%)	0.037	3333 (14%)	3242 (14%)	0.011
Benzodiazepines	3412 (13%)	4970 (13%)	0.019	3031 (13%)	2989 (13%)	0.005
Antiarrhythmics	3158 (12%)	5429 (14%)	0.046	2898 (12%)	2908 (12%)	0.001
Selective serotonin reuptake inhibitors	2892 (11%)	3179 (8%)	**0.107**	2404 (10%)	2452 (10%)	0.007
Tricyclic antidepressants	869 (3%)	944 (2%)	0.058	753 (3%)	713 (3%)	0.01
Topiramate	449 (2%)	483 (1%)	0.043	377 (2%)	371 (2%)	0.002
Pregabalin	313 (1%)	880 (2%)	0.078	309 (1%)	326 (1%)	0.006
Emergency department visit	3482 (14%)	9727 (25%)	**0.288**	3440 (15%)	3299 (14%)	0.017
Inpatient visit	919 (4%)	1146 (3%)	0.037	781 (3%)	750 (3%)	0.007

Note: This table includes baseline patient characteristics before and after propensity matching in this retrospective cohort study.
Asterisks (*) indicates the unmatched variable in our matching model. SMD values in bold indicate asymmetry between the cohorts, indicating by values greater than 0.1.
[a] standard deviation (SD)
[b] Non-steroidal anti-inflammatory drug (NSAID)
[c] standardized mean difference (SMD)

inpatient visits ($p < 0.001$; RR: 1.38 [1.29, 1.48]; ARI: 2.1%; 7.5% versus 5.4%) through one year after first-time diagnosis of CR (Table 2; Fig. 2). However, there was no statistically significant difference in risk of ED visits ($p = 0.638$; RR: 0.989 [0.943, 1.037]; 12.8% versus 12.9%). Finally, 47.7% of patients in the gabapentin cohort received at least one additional prescription of gabapentin after the date of diagnosis for CR.

4. Discussion

Our study, the largest study to date on this topic, examined safety and health service utilization markers in opioid-naïve adults prescribed either gabapentin or NSAIDs at the time of a new diagnosis of CR. We found that gabapentin was associated with increased risks of opioid prescription, ADEs, and markers of healthcare utilization over one year, highlighting the need to re-evaluate its use in management of CR. However, given that we could not exclude confounding by indication due to the unknown severity of first-time CR despite our rigorous study methodology and propensity matching strategy, these results should be taken with caution. The findings of this exploratory study should spur future research rather than sit as a final voice on this subject.

In our study, membership in the gabapentin cohort was associated with statistically significant increased opioid-related risks compared to the NSAID cohort, including a higher risk of oral opioid prescriptions (ARI = 4.9%), opioid-related ADEs (ARI = 0.2%), and new opioid-related disorders (ARI = 0.3%) within one year of CR diagnosis. Notably, 14.3% of patients in the gabapentin cohort received a co-prescription of opioids on the date of CR diagnosis. Co-prescription of gabapentin and opioids could partly explain our observed increased risk of adverse outcomes and utilization due to drug-drug interactions. First, opioid-mediated gastrointestinal slowing increases systematic gabapentin levels, thereby raising the risk of gabapentin-specific events such as respiratory depression [43], as seen in our study for acute respiratory

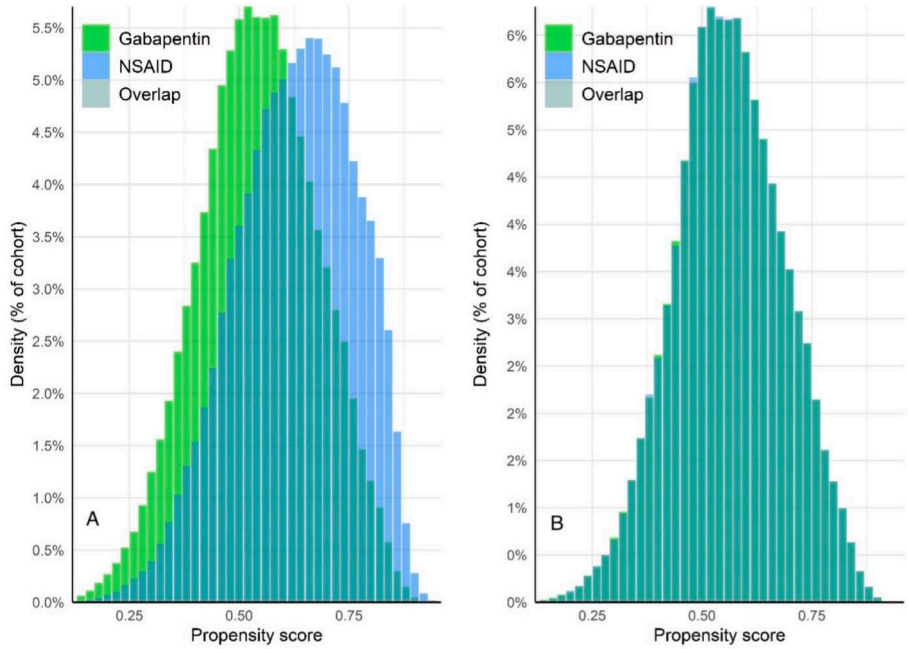

Fig. 1. This is a propensity score density graph. Shown are the gabapentin cohort (green) and the NSAID (blue) cohort with overlap shown in blue-green both before (A) and after (B) propensity matching. Following matching, the region of blue-green overlap predominates indicating adequate matching of covariates.

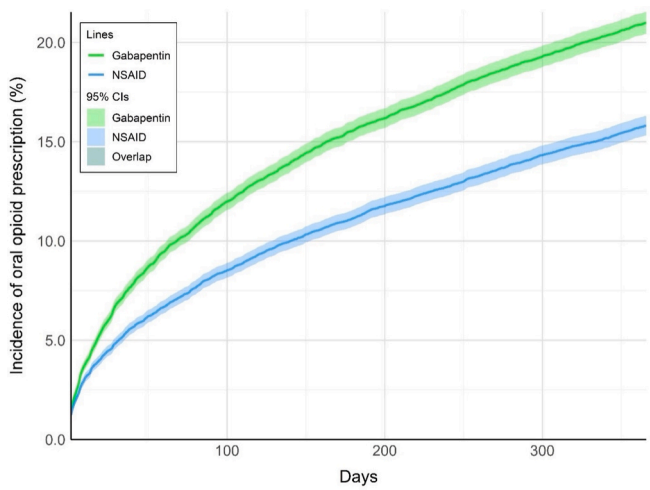

Fig. 2. This figure illustrates cumulative incidences of oral opioid prescription for patients in the gabapentin cohort (green) and the NSAID cohort (blue) with overlap (blue-green). Shaded areas represent 95% confidence intervals.

failure (ARI = 0.3%). Second, the central nervous system depressing effects of both medications may compound and further increase the risk of respiratory depression [23]. Third, gabapentin increases the effect of opioids and has previously been associated with opioid-related ADEs, such as overdose [43]. Clinicians should be aware that the Centers for Disease Control and Prevention (CDC) warns about the co-prescription of gabapentin and opioids [44]. In light of the opioid epidemic, these findings are concerning and likely do not represent ideal prescription practices. Future studies should examine the clinical significance of these prescription patterns and seek to implement harm reduction strategies when appropriate.

Additionally, our results indicated that gabapentin prescription was also associated with statistically significant increased likelihood of procedures and surgeries directed towards the cervical spine, including a higher risk of ACDF (ARI = 1.2%), PCF (ARI = 0.3%), and corticosteroid injection (ARI = 3.8%). The exact mechanism(s) underlying these associations is unclear. However, as a previous clinical trial found no pain relief advantage for gabapentin over epidural steroid injection [21], a lack of efficacy of the medication for neuropathic CR pain could explain care escalation [21], as patients or clinicians may seek out other effective treatments. It also remains possible that various detriments related to co-prescription of gabapentin and opioids could worsen the clinical picture and prompt surgical referrals or injections. Furthermore, our observed increase in ADEs and health service utilization among those prescribed gabapentin could also reflect systematic care factors. In this sense, factors associated with prescribing gabapentin could relate to clinician type, hospital setting, or region of practice [45]. While these or other non-clinical factors are outside of the scope of the present study to examine, they could mediate the relationship between gabapentin prescription and the observed outcomes. Alternatively, our findings could relate to patients obtaining prescriptions from multiple providers, coupled with gaps in tracking medications and potential interactions across health systems [46]. It is also possible that clinicians overprescribing gabapentin are also overutilizing other services, such as cervical imaging. Future studies examining real-world prescribing patterns, including more granular data such as prescribing provider specialty and initial healthcare setting to better clarify these care pathways and persuade improvement.

In the present study, patients who were prescribed gabapentin had no significant increase in risk of falls, dizziness, or somnolence, though these outcomes were slightly more frequent compared to the NSAID cohort. This may reflect type I error and underpowering, as our study relied on ICD-10 codes rather than patient surveys and events such as falls may be under recorded. Given the low incidence of events, null findings related to these outcomes should be interpreted cautiously. Additionally, our analysis of healthcare visits showed mixed findings, with a null result for ED visits alongside a statistically significant increase in risk of inpatient visits among those prescribed gabapentin, corroborating the results of Peckham et al. (2018) [26]. While the mechanisms underlying the conflicting ED/inpatient visit pattern are unclear, it is possible that adverse events such as respiratory failure or opioid-related ADEs, which were more frequent in the gabapentin cohort, could prompt hospital admission versus ED visits alone.

Table 2
All safety/adverse drug events and health service utilization outcomes for both cohorts.

Outcome	Gabapentin cohort	NSAID cohort
Safety and Adverse Drug Events		
Oral opioid prescription, N (%)	4,479 (19.2%)	3,354 (14.3%)
RR[a] ([95% CI[b]]; *P*)	1.34 ([1.28, 1.39]; *P* < *0.001*)	(reference)
Opioid-related disorder, N (%)	89 (0.4%)	31 (0.1%)
RR ([95% CI]; *P*)	2.87 ([1.91, 4.32]; *P* < *0.001*)	(reference)
Opioid-related adverse drug event, N (%)	65 (0.3%)	28 (0.1%)
RR ([95% CI]; *P*)	2.32 ([1.49, 3.61]; *P* < *0.001*)	(reference)
Acute respiratory failure, N (%)	133 (0.6%)	74 (0.3%)
RR ([95% CI]; *P*)	1.80 ([1.35, 2.39]; *P* < *0.001*)	(reference)
Dizziness, N (%)	949 (4.1%)	882 (3.8%)
RR ([95% CI]; *P*)	1.08 ([0.98, 1.18]; *P* = 0.110)	(reference)
Somnolence or disorientation, N (%)	114 (0.5%)	90 (0.4%)
RR ([95% CI]; *P*)	1.27 ([0.96, 1.67]; *P* = 0.092)	(reference)
Fall, N (%)	404 (1.7%)	360 (1.5%)
RR ([95% CI]; *P*)	1.12 ([0.98, 1.29]; *P* = 0.108)	(reference)
Health Service Utilization		
Anterior cervical discectomy and fusion, N (%)	529 (2.3%)	252 (1.1%)
RR ([95% CI]; *P*)	2.10 ([1.81, 2.44]; *P* < *0.001*)	(reference)
Posterior cervical fusion, N (%)	100 (0.4%)	31 (0.1%)
RR ([95% CI]; *P*)	3.23 ([2.16, 4.82]; *P* < *0.001*)	(reference)
Cervical corticosteroid injection, N (%)	1,473 (6.3%)	584 (2.5%)
RR ([95% CI]; *P*)	2.52 ([2.30, 2.77]; *P* < *0.001*)	(reference)
Emergency department visit, N (%)	2,985 (12.8%)	3,019 (12.9%)
RR ([95% CI]; *P*)	0.99 ([0.94, 1.04]; *P* = 0.638)	(reference)
Inpatient visit, N (%)	1,755 (7.5%)	1,271 (5.4%)
RR ([95% CI]; *P*)	1.38 ([1.29, 1.48]; *P* < *0.001*)	(reference)
Cervical spine radiograph, N (%)	2,813 (12.0%)	2,329 (10.0%)
RR ([95% CI]; *P*)	1.21 ([1.15, 1.27]; *P* < *0.001*)	(reference)
Cervical spine magnetic resonance imaging, N (%)	4,866 (20.8%)	3,395 (14.5%)
RR ([95% CI]; *P*)	1.43 ([1.38, 1.49]; *P* < *0.001*)	(reference)

Notes: This table includes all safety/adverse drug events and health service utilization outcomes for this retrospective cohort study for both cohorts.
[a] Risk ratio (RR).
[b] 95% confidence intervals (95% CI).

The overall clinical significance of our findings can be interpreted according to the outcome type using context of RR magnitude, risk difference, and incidence estimates [47,48]. Larger-magnitude RR effect estimates especially exceeding two are less likely explained by confounding when statistically significant [48]. Accordingly, these outcomes included the observed increased risk of ACDF, opioid-related ADEs, cervical corticosteroid injection, opioid-related disorder, and PCF for gabapentin as compared to NSAIDs. The summation of these data cautiously raise concerns and seem to parallel existing skepticism in the current literature [5] to avoid gabapentin prescription for CR, aligning with the aforementioned lack of explicit guideline recommendations [14]. Our findings suggest any potential benefits of gabapentin for CR should be carefully weighed against its risks. Given the already limited evidence supporting gabapentinoids for radicular syndromes [21], our study highlights additional potential disadvantages of its use for CR. Notably, we did not find any safety or utilization advantages for its prescription compared to NSAIDs among opioid-naïve adults with CR. Overall, such findings appear to support previous guidelines and recommendations to avoid gabapentin prescription for CR while opting for patient education, nonpharmacologic care such as exercise, or other medications such as NSAIDs [5].

5. Strengths and Limitations

Strengths of this study included being the largest study examining gabapentin prescription for CR, a large adequately powered sample size with extensive propensity matching strategy, a multidisciplinary research team, and a robust assessment of multiple outcomes. Our study cohort demographics (mean age: 53 years; 57% female) align with those expected for CR [49], supporting our study's validity and generalizability.

Certain limitations should be considered. This study was retrospective, and therefore suffered from biases inherent to observational study designs. For example, it is possible that selection bias and misclassification bias are still present, particularly in the absence of a formal diagnostic look-back period, despite our best attempts via rigorous inclusion and exclusion criteria. Notably, most patients had substantial longitudinal data available (83% of the NSAID cohort and 84% of the gabapentin cohort, ≥24 months), which provides contextual reassurance but does not substitute for a formal look-back period, warranting acknowledgement. A central limitation of this study is confounding by indication, as CR can represent a clinical spectrum and we were unable to ensure homogenous disability or pain severity across cohorts: a constraint that may persist despite matching via markers of healthcare utilization. This raises substantial concerns that prescription decisions, and thus observed associations, may have been driven by underlying disease severity. Thus, future studies should categorize patients by pain severity, functional status, neurological deficit, or specific imaging findings to create a more homogenous patient cohort and simultaneously examine clinical measures of pain or disability as outcomes, which this study cannot assess due to data limitations. It is also important to remember that association is not the same as causation, especially given the biases inherent to a retrospective study such as the current study. This study also cannot account for NSAID or gabapentin dosage variability (e.g., pill count, type, or strength), treatment duration, or conduct subgroup analyses by dose. While opioid and gabapentin co-prescription at the index date was evident, subsequent prescriptions may have occurred at different timepoints. It is also important to note that these prescription practices may serve only as initial treatment strategy, rather than continuous therapy, for CR. There may also be underreporting of certain outcomes like falls in ICD-10 codes due to reliance on structured data without free-text access. There may also be variation in individual physician or specialty prescribing practices, which we cannot account for given data constraints. There was an inability to determine if opioid-related adverse events were fatal or whether patients adhered to gabapentin as prescribed versus misused it. Future research should investigate safety/ADEs and health service utilization outcomes in relation to prescription parameters (e.g. dosage and frequency) while controlling for severity of CR in adult patients. Findings may not be generalizable outside of the US, considering differences in prescription patterns or pharmacologic approaches to CR.

5.1. Conclusion

Gabapentin prescription for a new diagnosis of CR in opioid-naive adults is associated with an increased risk of oral opioid prescription, multiple opioid-related ADEs, and new opioid-related disorders as well as increased markers of health service utilization such as increased risk of cervical spine fusion, corticosteroid injection, and cervical spine

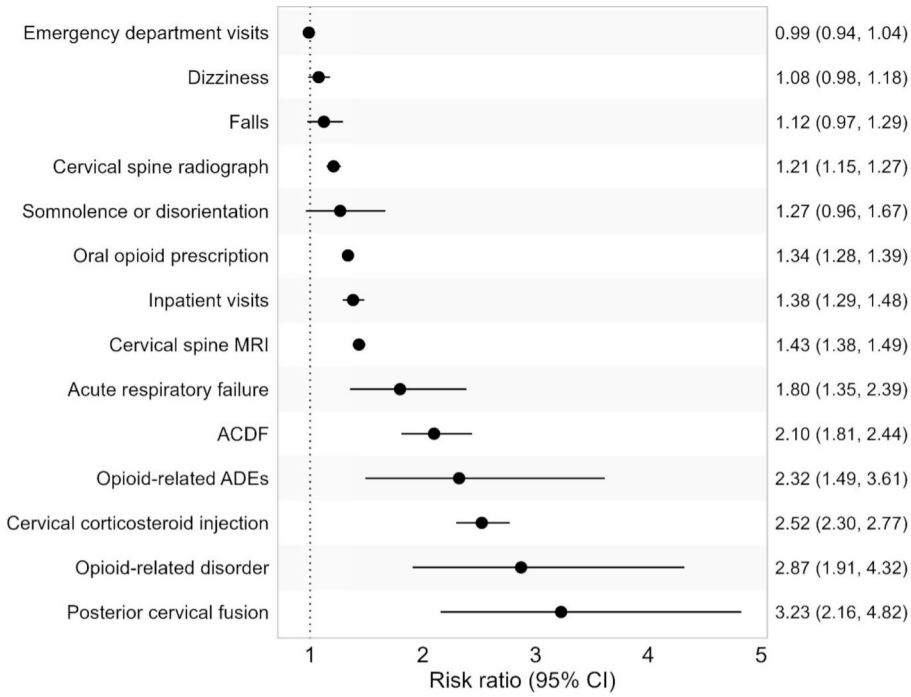

Fig. 3. This is a forest plot displaying the safety/adverse drug events and health service utilization outcomes. Risk ratios exceeding one indicate events more likely to occur in the gabapentin cohort relative to the non-steroidal anti-inflammatory drug (NSAID) cohort. Abbreviations: ADEs; adverse drug events, ACDF; anterior cervical discectomy and fusion, MRI; magnetic resonance imaging. Outcomes (left) are shown in ascending order based on risk ratio point estimates (right) with 95% confidence intervals.

imaging, through one year after diagnosis of CR. Importantly, none of the outcomes favored gabapentin. Furthermore, about 14% of patients prescribed gabapentin were co-prescribed opioids at the time of diagnosis of CR, highlighting gaps in clinical practice guideline adherence. In light of the concern regarding off-label gabapentin prescription for musculoskeletal conditions, this study suggests that prescribing practices for gabapentin warrant greater reconsideration.

CRediT authorship contribution statement

Anthony N. Baumann: Writing – original draft, Supervision, Investigation, Formal analysis, Conceptualization. **Robert J. Trager:** Writing – original draft, Validation, Supervision, Conceptualization. **Dalton McGlamery:** Writing – Review and Editing, Project Administration. **Shahabeddin Yazdanpanah:** Writing – review & editing, Project administration. **Keegan T. Conry:** Supervision, Project administration. **Gordon Preston:** Supervision, Project administration. **Jacob C. Hoffmann:** Supervision, Resources, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethical Review Committee Statement.

This study included de-identified data from TriNetX which is considered "Not Human Subjects". Therefore, the study is exempt from IRB approval and patient consent requirements based on University Hospitals' standard operating procedures.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jocn.2026.111896.

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